

8	(a)	<u>larger energy gap</u> between conduction band and lower orbitals than standard silicon	M1
		resulting in a larger energy / higher frequency photons	A1
	(b)	(i)	C1
		wavelength = 595 nm	
		energy = $(6.63 \times 10^{-34})(3.00 \times 10^8) / (595 \times 10^{-9}) (= 3.343 \times 10^{-19} \text{ J})$	M1
		energy = $3.343 \times 10^{-19} / 1.60 \times 10^{-19} = 2.09 \text{ eV}$ or 2.1 eV	A1
		(ii)	B1
		1. insufficient energy for electron to promote / excite to appropriate energy level	
		2. energy of the photons greater / wider gap between the conduction band and the lower orbitals than red LED	B1
		so higher energy per unit charge is required	
	(c)	(i)	B1
		point correctly plotted	
		(ii)	B1
		even distribution of points on either side of the line along the full length	
		(iii)	M1
		correct method to compute gradient i.e. $\Delta y / \Delta x$ and coordinates are read accurate to half the smallest square	
		gradient calculated correctly (most line will give a gradient of about -1.7 to -1.8)	A1
	(d)	(i)	M1
		<i>either</i> obtain $\lg V_F$ from graph and $V_F = 10^{\text{value}}$ (e.g. $\lg(520) = 2.716$, $\lg V_F = 0.435$) <i>or</i> calculate k from y-intercept and substitute 520 nm into $V_F = k \langle \lambda \rangle^n$	
		$V_F \approx 2.7 \text{ V}$ (final answer depends on the line of best fit and rounding in the intermediate calculations)	A1
		(ii)	M1
		p.d. across resistor = $4.5 - V_F$ from (d)(i) resistance = p.d. / 20 mA	
		resistance $\approx 90 \Omega$	A1
	(e)	(i)	M1
		$v = c / n = 3.00 \times 10^8 / 4.24$	
		$v = 7.08 \times 10^7 \text{ m s}^{-1}$	A0
		(ii)	C1
		$n_1 \sin \theta_1 = n_2 \sin \theta_2 \Rightarrow 4.24 \sin \theta = (1.00) \sin 90^\circ$ $\theta = 13.6^\circ$	A1
		(iii)	M1
		coating increase critical angle so more photons emerge / allows photons with larger angles to the normal of the surface of diode (that are otherwise trap in diode) to emerge	
		these photons (with those emerging perpendicularly) are then reflected by plastic bulb and concentrated in forward direction	A1
		so, more photons released for given energy input	
	(f)	ratio = $(900 / 9) / (840 / 60)$	M1
		= 7.1	A1